Actionable Real-Time Modeling of Surgical Team Dynamics via Time-Expanded Interaction Graphs

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Operating Rooms Insights

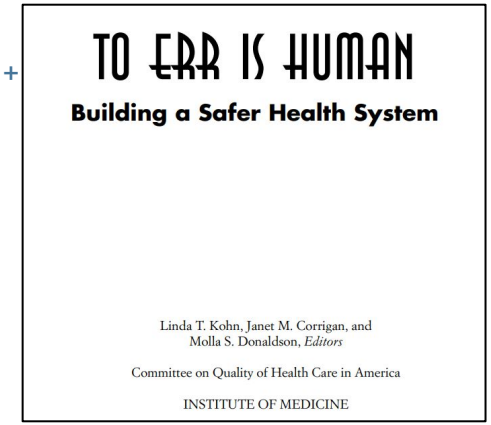


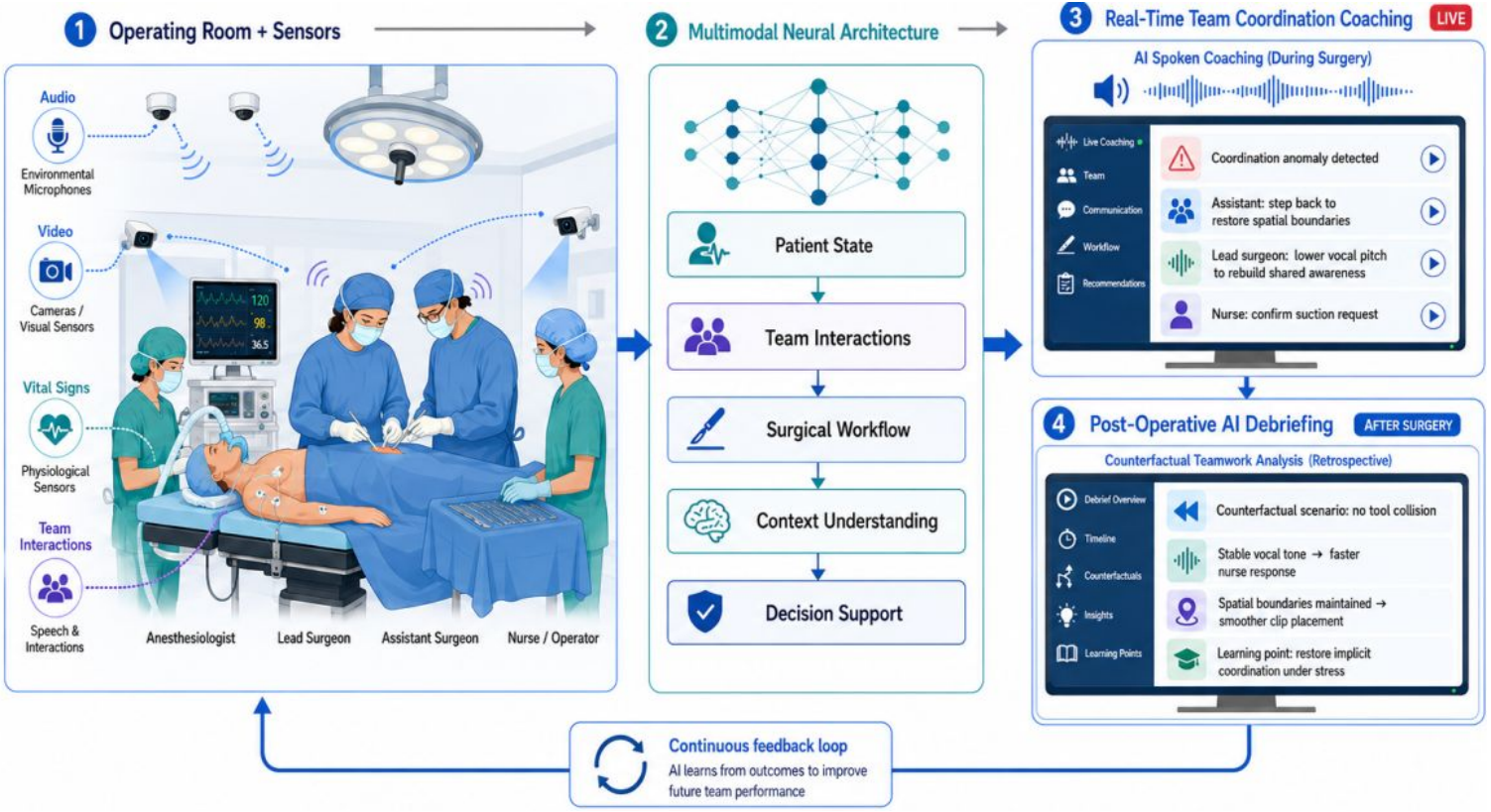
Figure: (Brown, 2001)

- Medical errors are the **third leading cause of death** in the US
- One of the main reasons for medical errors is **poor teamwork in the OR**

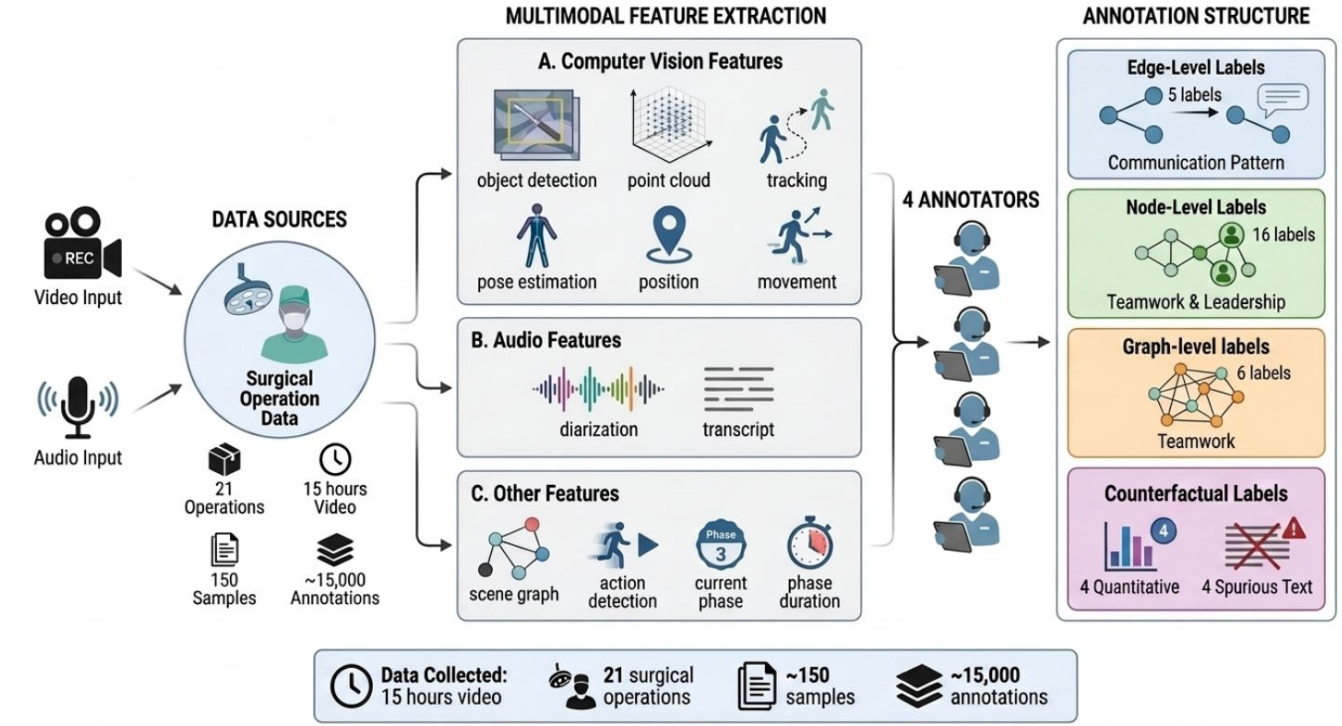
Missing components for automated team analysis in literature:

1. Lack of datasets capturing interactions in the OR and proper data annotations for team data
2. Lack of architectures modeling team behavior generating accurate predictions
3. Design **agents able to provide feedback to OR team**

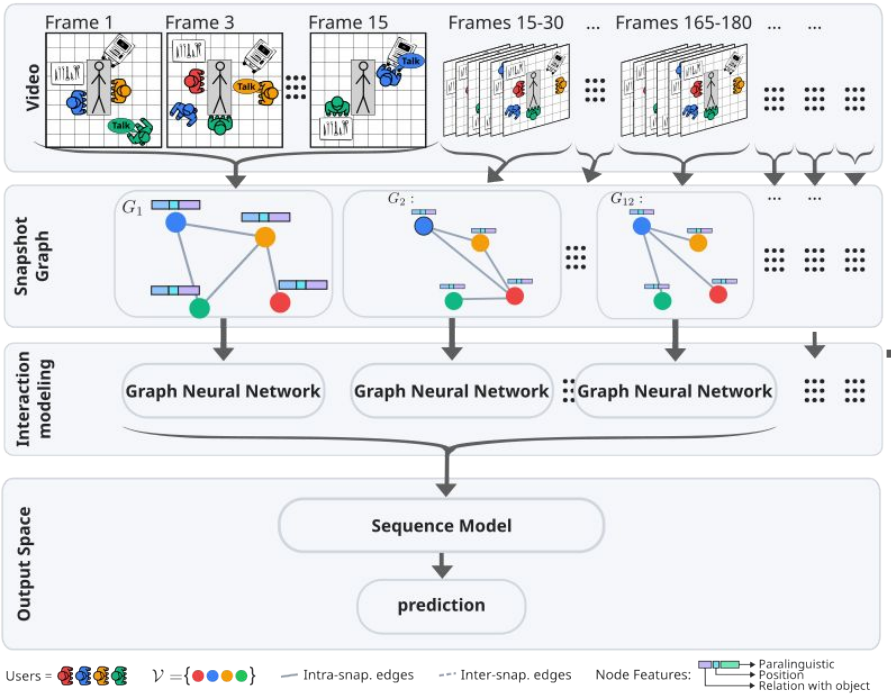
Operating Rooms Design

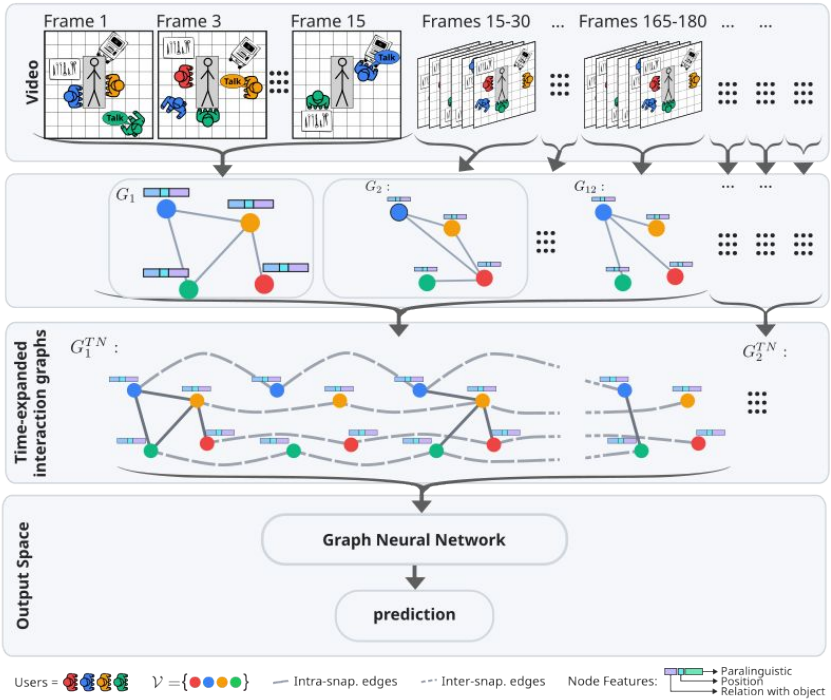


1. Operating Room Data



2. From Temporal To Time-Expanded GNN





3. Feedback - Evaluation

1) Adding Communication



Situation. The nurse asks a question, but the leader is distracted and does not respond, leading to silence and low communication.

Counterfactual. Add edges

Interpretation. Reduce disengagement by increasing leader responsiveness.

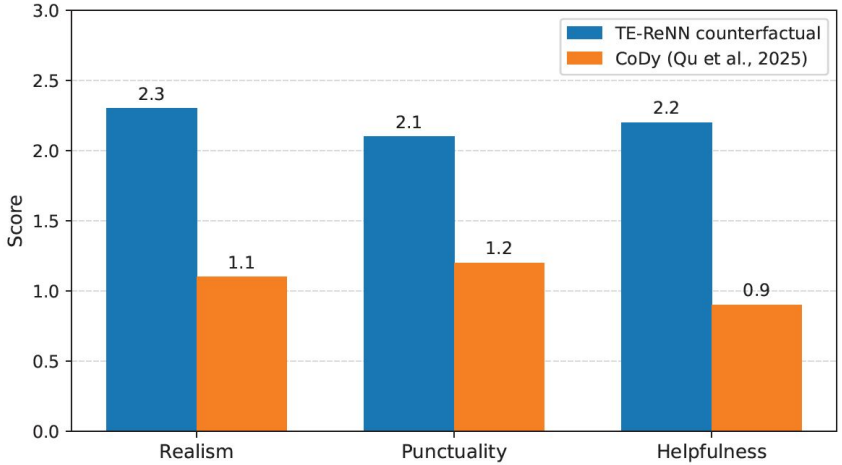
4) Aggressive Communication



Situation. A rude response from the leader to a nurse's question reduces team cooperation.

Counterfactual. Increase "Control" (low → high).

Interpretation. Adopt a "Calm Leader" style instead of "Dominant–Aggressive".



Thank you for the attention!

Boosting Team Modeling through Tempo–Relational Representation Learning, Cognitive Communication, 2025

Towards Actionable Surgical Team Dynamics: from Teamwork to Counterfactual Annotations, Human-AI Complementarity and Alignment, 2026

Actionable Real-Time Modeling of Surgical Team Dynamics via Time-Expanded Interaction Graphs , Human-Hybrid Artificial Intelligence, 2026

From Multimodal Observation to Interpretable Suggestions: Counterfactual Time-Expanded Relational Modeling of Surgical Teams, ACM Multimedia, 2026